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REGRESSION

Input / Training Data $\rightarrow \{x_1, \ldots, x_n\}$ $\quad x_i \in \mathbb{R}^d$
Output $\rightarrow \{y_1, \ldots, y_n\}$ $\quad y_i \in \mathbb{R}$

Goal: Learn $h : \mathbb{R}^d \rightarrow \mathbb{R}$.

How to measure goodness of h?

$$\text{error}(h) = \sum_{i=1}^{n} (h(x_i) - y_i)^2$$

Linear Structure

$$\mathcal{H}_{linear} = h_w : \mathbb{R}^d \rightarrow \mathbb{R}$$

such that, $h_w(x) = w^T x \quad \forall\, w \in \mathbb{R}^d$

Modified Goal

$$\min_{h_w \in \mathcal{H}_{linear}} \sum_{i=1}^{n} (h_w(x_i) - y_i)^2$$

or

$$\min_{w \in \mathbb{R}^d} \sum_{i=1}^{n} (w^T x_i - y_i)^2$$

- If a linear regression model achieves zero training error, all data points lie on a hyperplane in the $(d+1)$ dimensional space.

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Objective function can be rewritten as:

$$\min_{w \in \mathbb{R}^d} \|X^T w - y\|_2^2$$

$$X^T = \begin{bmatrix} — & x_1 & — \\ — & x_2 & — \\ & \vdots & \\ — & x_n & — \end{bmatrix}_{n \times d} \quad w = \begin{bmatrix} | \\ w \\ | \end{bmatrix}_{d \times 1} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}$$

$$\min_{w \in \mathbb{R}^d} (X^T w - y)^T (X^T w - y)$$

↑

Unconstrained Quadratic optimisation problem

Solution

$$f(w) = (X^T w - y)^T (X^T w - y)$$

$$\nabla f(w) = 2(XX^T)w - 2(Xy)$$

Solution satisfies $(XX^T)w^* = Xy$

$$w^* = (XX^T)^+ Xy$$

$+ \rightarrow$ pseudo inverse

Geometric Interpretation

$X^T w^*$ is the projection of the labels onto the subspace spanned by the features.

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Weight Vector / Closed form solution of LR

$$w = (XX^T)^{-1} X y$$

Kernel function - Gaussian

$$k(x_i, x_j) = \exp\left(\frac{-\|x_i - x_j\|^2}{2(\sigma^2)}\right)$$

Prediction

$$\sum_{i=1}^{n} k(x_i, x_{test})\, \alpha_i$$

Gradient Descent

$$w' = w^0 - \eta\,[2XX^T w^0 - 2Xy]$$

Kernel Function - Polynomial degree 2

$$k(x_i, x_j) = (x_i^T x_j + 1)^2$$